



Using Relational Training to Improve Performance During Acceptance and Commitment Training Sessions

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Abstract

The current study investigated the effect of relational training on improving engagement during acceptance and commitment training (ACT) sessions that aimed to reduce maladaptive behavior. Three female children diagnosed with autism participated in the current research. The relational training protocol involved eight programs designed to improve participants' ability to engage in relational responding. A concurrent multiple-baseline across-participants design was used, and participants' maladaptive behavior during ACT sessions was recorded. Results indicated a decrease in maladaptive behavior for one of the three participants, with a medium to large treatment effect. A secondary measure also showed improvement in all three participants' performance during ACT sessions. The percentage of nonoverlapping data and Cohen's *d* suggest a small to moderate treatment effect. Implications for implementing ACT with individuals diagnosed with autism are discussed.

Keywords Acceptance and commitment training (ACT) · Relational training · Autism · ACT Quantitative Analysis Scale (AQAS) · PEAK · AIM

The field of applied behavior analysis (ABA) has seen a growing interest in implementing language-based interventions with children diagnosed with autism in recent years. A number of studies have been published that provide behavior-analytic

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Research Highlights

- Individuals diagnosed with autism often demonstrate deficits in language and abilities to engage in relational responding, which may negatively impact their ability to benefit from acceptance and commitment training (ACT).
- ACT successfully decreased maladaptive behavior for one of the three participants.
- Relational training was successful in improving all participants' performance in ACT sessions.
- Practitioners should incorporate relational training to facilitate ACT-based instructions.

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accounts of these approaches and demonstrate their effectiveness (Wilson, Sandoz, Flynn, et al., 2010; Wilson, Sandoz, Kitchens, & Roberts, 2010). As practitioners often find themselves in situations with limited options for addressing concerns related to private events or rigid rule-governed behaviors, recent development in acceptance and commitment training (ACT) is at the center of this movement of expanding the scope of behavior-analytic practices. This is further evidenced by the Behavior Analyst Certification Board (2020) officially recognizing the practice of ACT as an ABA subspecialty.

Rooted in relational frame theory, ACT targets six different processes conceptualized as the psychological flexibility model: present-moment awareness, acceptance, defusion, self-as-context, values, and committed action. Each of these six processes contributes to overall psychological flexibility and can be targeted using different metaphors and activities (Fletcher & Hayes, 2005). Fletcher and Hayes (2005) provided a concise definition for all six processes of the model. *Acceptance* can be conceptualized as the ability to experience private events in the moment without attempts to change their frequency or form. *Defusion* is the ability to separate oneself from their private events by altering the function of that thought. *Present-moment awareness* describes the ongoing, nonjudgmental contact with stimuli in the current environment. *Self-as-context* describes the ability to contact the

transcendent self that emerges as the result of deictic responding. *Values* describe the chosen life direction and can be conceptualized as an abstract, delayed conditional reinforcer that functions as the motivating operation. Last but not least, *committed action* refers to the behavior an individual engages in to service their values. Together, these six processes offer a powerful tool that clinicians can use to conceptualize interventions that ultimately produce socially significant changes. As a practitioner, targeting these processes can improve clients' social-emotional deficits, rigid rule following, and psychological and behavioral inflexibility (Brown & Hooper, 2009; Enoch & Dixon, 2017, 2019). For example, clinicians can improve clients' skills on acceptance by developing their behavior to tolerate aversive stimuli without avoidance. Clinicians can address the self-as-context process by practicing perspective-taking skills using deictic relations (e.g., I-you, now-then, here-there). Clinicians can also engage in value clarification exercises, so clients are less likely to satiate on one particular reinforcer.

Practitioners have demonstrated success in applying ACT in various settings related to autism treatment. Blackledge and Hayes (2006) showed that a 14-hr ACT workshop was effective in helping parents adjust to the difficulties in raising a child diagnosed with autism. Similarly, Hahs et al. (2019) demonstrated that parents of children diagnosed with autism who received ACT showed significant improvements in six of eight psychological flexibility measures. Yi and Dixon (2021) also demonstrated that a brief ACT component could effectively increase adherence to a telehealth ABA parent-training curriculum for caregivers of children diagnosed with autism. Recently, more research has begun to evaluate the utility of implementing ACT directly with individuals diagnosed with autism—in particular, regarding its effect on decreasing maladaptive behavior. For example, Szabo (2019) successfully decreased inflexible behaviors with three males diagnosed with autism. Singh et al. (2019) demonstrated the effectiveness of using ACT to decrease verbally and physically aggressive behavior with three adolescent males diagnosed with autism. Although a growing amount of research is advocating the use of ACT, practitioners working with individuals affected by autism must determine whether this technique is appropriate for their clients, as individuals diagnosed with autism often show delays in language, and language-based interventions such as ACT may present unique challenges.

For example, a common ACT activity used by clinicians is value clarification. From a behavior-analytic point of view, values can be conceptualized as abstract conditional motivating operations that will increase the likelihood of the client engaging in behavior that will lead them to access available reinforcers. However, these concepts are abstract, and the reinforcing consequence is often delayed. This could be challenging for individuals diagnosed with autism (e.g., the client may be able to identify that they want to work for playing outside with friends, but

they are unable to identify the value of friendship). The inability to identify an abstract and delayed reinforcer, or “values,” could make the client easier to satiate on a particular reinforcer. Additionally, clinicians may ask the client to identify behavior that confronts their values and those that do not. This requires the client to respond to frames of opposition, coordination, and didactic responding, which may require additional training.

Another example is the process of acceptance. Behavior-analytically, acceptance can be conceptualized as the repertoire of tolerating aversive stimuli without engaging in escape-maintained behavior. However, individuals diagnosed with autism often demonstrate a higher level of impulsivity and may not be able to tolerate the immediate aversiveness of a stimulus in exchange for a delayed appetitive consequence (Bradley & Isaacs, 2006). Moreover, such aversiveness might be caused by an indirect stimulus function established by a prior learning history (Roche & Barnes, 1997). For example, if a child was bitten by a dog and has established an equivalence class between the animal “dog” and the word “dog,” the child may engage in avoidant behavior in the presence of the word “dog.” Without the means to expand the equivalence class, or to establish appropriate contextual control, the alternative acceptance behavior may be difficult to maintain.

As ACT's theoretical and philosophical foundations are rooted in relational frame theory, one may assume that the individual must be able to readily engage in behavior accounted for by relational frame theory in order for ACT to be effective (Hayes, 2004). Hypothetically, if a child is unable to demonstrate mutual entailment, combinatorial entailment, or transformation of stimulus functions at all, then whatever maladaptive behavior the child engages in must be brought under stimulus control via a history of direct reinforcement. In this case, ACT's conceptualization of psychopathology, which claims that it evolves, in part, as the derived relations dominate other sources of behavior regulations (Fletcher & Hayes, 2005), may not hold true. Moreover, as ACT relies heavily on using metaphors and experiential exercises to influence each of the six processes in order to establish flexible behavior, whether repertoires and relations established in these activities can be applied in their own environment seems highly dependent on whether the individual can relate two relational networks. As Stewart et al. (2001) pointed out, the ability to understand metaphor requires an individual to establish a frame of coordination between two separate relational networks. Deficits in one's ability to engage in derived relational responding are detrimental to the individual's ability to effectively modify the target relational network and, in turn, may result in ineffective treatment. From this perspective, practitioners should pay special attention to the client's ability to engage in arbitrarily applicable relational responding, as verbally capable individuals may still have underlying deficits in derived relational responding (Sutton et al., [under review](#); Barnes-Holmes

et al., 2000; Dixon et al., 2015). This should especially be the case for practitioners working with clients diagnosed with autism, as they often demonstrate difficulties in relational framing (Devany et al., 1986; Kent et al., 2017).

The field of ABA has seen a growing interest in identifying and addressing deficits of derived relational responding with individuals diagnosed with autism (Belisle et al., 2020). Different assessments have been developed, along with treatment protocols that explicitly target clients' ability to engage in relational framing (Dixon, 2016, 2019). For example, Dixon et al. (2017) generated derived categorical responding with three participants diagnosed with autism using a match-to-sample procedure. Belisle et al. (2016) successfully improved the ability to engage in single-reversal "I–You" deictic responding with three participants with autism. In another study by Dixon et al. (2019), participants demonstrated significant gains in IQ following 18 weeks of relational training. However, the majority of these studies only focus on the frame of coordination. Other than the one study that included a randomized controlled trial that evaluated the overall effect of relational training (Dixon et al., 2019), most of studies evaluated relational training procedures as a stand-alone intervention to develop a specific skill, not as a treatment package to improve overall responding (Belisle et al., 2020). No study to date has examined the effect of relational training on ACT interventions. This could be especially important to individuals diagnosed with autism, as their deficits in relational responding might undermine ACT's outcome.

Therefore, the current study investigated the effect of using relational training on facilitating participation in ACT with individuals diagnosed with autism in order to decrease maladaptive behavior. Participants received ACT along with relational training based on a standardized relational training curriculum (Dixon, 2016). We evaluated to what extent ACT decreased their maladaptive behavior. Furthermore, we examined whether the introduction of the relational training protocol improved ACT's treatment effect. We used a secondary measure to capture participants' responding during ACT sessions, in terms of the precision, scope, and depth of their verbal behavior. Limitations and implications for practitioners using ACT with individuals diagnosed with autism are discussed.

Method

Participants and Settings

Three children diagnosed with autism receiving ABA services in a clinic located at a midwestern university participated in the current study. All three participants had no prior exposure to ACT. The first participant, Briana, was a 10-year-old girl diagnosed with autism. Before participating in the current

study, Briana had been receiving ABA services since she was 18 months old and had been receiving received ABA services at this clinic for 2 years. At the time of the study, she was taking Focalin and clonidine. The second participant, Josie, was a 10-year-old girl diagnosed with autism. Before participating in the current study, Josie had been receiving ABA services at the clinic for 1 year. She did not take any medication. The third participant, Tina, was a 9-year-old girl diagnosed with autism and oppositional defiant disorder. Before participating in the current study, she had been receiving ABA services at this clinic for 2 years. At the time of the study, Tina was taking Concerta.

All sessions were conducted in two locations. The first location was one of four identical therapy rooms within the clinic. All therapy rooms were minimally decorated, and each therapy room had two to three chairs with a rectangular child-size table. In addition to the therapy room, Tina completed seven sessions in an office that contained a round table and two to six chairs. The office was also minimally decorated. Each of the participants had two sessions per week, and each session lasted 2 hr. Each session began with 30–60 min of ACT-based instruction. After the ACT instruction, participants either returned to their regular ABA session or received the relational training curriculum depending on their phases in the study. Participants spent 20 weeks in the current study. Sessions were conducted by three 2nd-year master's-level students enrolled in a behavior analysis program.

Intervention Protocol

ACT

The ACT protocol used in the current study was adapted from the Accept Identify Move curriculum developed by Dixon and Paliliunas (2018). The protocol consisted of 31 experiential activities that targeted all six processes of the psychological flexibility model. At the beginning of each session, the participants were asked to recall what ACT stood for, to list the six processes, and to say whether they could remember what each process meant. They were then shown a picture of an hexaflex, which illustrated the psychological flexibility model, and were asked to define each process. If they could not, answers were provided to them. Then, the clinician guided the participant through the experiential activity and led the discussion to ensure that participants understood the material. Table 1 provides a summary of this protocol. The detailed protocol is available upon request from the corresponding author.

Relational Training

The relational training curriculum used in the current study included eight programs adapted from the Promoting the

Table 1 Summary of the acceptance and commitment training protocol

Intervention protocol	Hexaflex component	Experiential activity
Day 2: Saying Yes	Acceptance	Recognizing thoughts is like recognizing songs. Initially, you may not like the song, but over time, it becomes better.
Day 3: Cutting the String	Defusion	We can allow our thoughts to float away like balloons in the air; they do not disappear but become less powerful.
Day 4: Flat Tire	Committed action	Like a road trip, sometimes we encounter a flat tire or event that slows us down. Rather than giving up, we fix our flat tire and continue our journey.
Day 5: The Long Jump	Values	Values are like the finish line at the end of a long jump. Sometimes you must work harder or jump farther to get closer to things that are important to you.
Day 6: Chameleon	Self-as-context	Similar to a chameleon, we think and behave differently in different situations.
Day 7: You've Got a Friend . . . in You	Acceptance	Think about a time you messed up. Rather than judging yourself, think of thoughts you could say to yourself as a supportive friend.
Day 8: Thought Bubbles	Defusion	We experience good thoughts and thoughts that might cause us pain. These thoughts are like bubbles that come and go and float by without getting stuck.
Day 9: My Totem	Values	When life gets busy, it is easy to lose focus on our values. Assign an animal to represent your values, and make a totem pole as a reminder of those values.
Day 11: Life Ladder	Committed action	Working toward your values can be a lot like climbing a ladder. Each step can be seen as one goal or action you take to get closer to your values.
Day 12: Centered Chaos	Present-moment awareness	Mindfulness is more than just sitting quietly free from distractions. It is important to practice mindful awareness when the environment is busy and noisy.
Day 14: A Picture of Struggle	Acceptance	We have thoughts and feelings we struggle with. Imagine your struggles as colors, and paint a picture without them; your picture may be incomplete.
Day 22: Writer, Producer, Director!	Self-as-context	We may step back to watch our life as it occurs, forgetting that we play a major role in it. In life, we get to write, produce, and direct our movie.
Day 26: Walking to Cloud Mine	Values	Values are like clouds. From far away, they may seem small but become bigger as you get closer. You will never reach the cloud, but it will become clearer the closer you get.
Day 27: Just One Thing	Committed action	It can feel overwhelming thinking about all the committed actions needed to reach your values. Pick one committed action you can do each day.
Day 28: Concentration	Present-moment awareness	Rather than just passing through the day, spend a few minutes observing something right in front of you. See if you notice things you have not noticed before.
Day 29: From Miserable to Magical	Defusion	Think of something that made you really sad and then something that made you laugh or smile. Even painful or upsetting feelings are usually temporary.
Day 30: Eye, Observing	Self-as-context	Blurry vision can cause us to see things differently than what they really are. Mindfulness helps us clear our mind and notice what is going on around us.
Day 31: Are You Willing to Enjoy It?	Acceptance	It is easy to get stuck thinking about the things we want in life such that we can become unwilling to experience or enjoy events right in front of us.
Day 32: The Finish Line	Committed action	In a race, we encounter barriers that get in our way. When we face these barriers, we can choose to fight against them or address the barrier and move forward.
Day 33: You Are Not the Thinker	Self-as-context	When we have thoughts, we have a choice: to fight the thoughts or to be willing to fully experience those thoughts.
Day 34: Simply Grateful	Present-moment awareness	People get caught up in their schedules and activities and forget to appreciate the little things in life. Focusing on too many things makes it easy to forget to show gratitude for the things we care about.
Day 39: Opposites	Self-as-context	We use words to describe ourselves such as "smart," "silly," "bad," etc. Consider whether these words are who we really are or just words describing the things we do.
Day 40: Who CARE-acters	Values	There are people in our lives we look up to. Consider why you look up to these people; perhaps it is because they represent something you value.
Day 42: The Point of It All	Values	Although you have many values in your life, it can sometimes be difficult to stick to those values even when doing so will pay off in the long run.
Day 46: Let It Rain	Acceptance	Getting stuck in the rain without an umbrella can be a bummer. Instead of standing in the rain, letting it ruin your day, you can accept it, let it go, and move on.
Day 50: One With Nature	Present-moment awareness	Next time you go outside, focus on the present moment by focusing on the sights, sounds, and smells around you.
Day 69: Bad Kid	Self-as-context	You may have certain thoughts about yourself. However, it is important to notice the difference between the real you and the evaluations of yourself.
Day 73: Asteroids	Acceptance	Bad thoughts and feelings can feel like they are speeding toward you like an asteroid flying toward a spaceship. You can either avoid them or accept them.
Day 81: Tournament of Values	Values	You may find that you value many things in your life. Sometimes, it is helpful to prioritize those values to help you make decisions.

Emergence of Advanced Knowledge (PEAK) curriculum developed by Dixon (2016). All relational trainings were delivered via a tabletop discrete-trial training format. Prompting was used to facilitate the correct response. These programs corresponded to different processes outlined in the psychological flexibility model and included “Committed Action” and “Values, Goals, and Actions” targeting the committed action process; “Defusion” and “Defusion With Art” targeting the defusion process; “Self-as-Context” targeting the self-as-context process; “Values Perspectives” targeting values; “Acceptance” targeting the acceptance process; and “I Contact” targeting present-moment awareness. Six major relational frames were targeted in this treatment package: correspondence, opposition, distinction, hierarchical, comparison, and deictic. Each program consisted of multiple steps, and training was conducted in various 10-trial blocks. Participants never received feedback on some of the steps. This allowed the clinician to assess whether the participant could derivate relations with novel stimuli without a previous history of reinforcement. Alphanumeric notation provided in what follows delineates the location within the PEAK curriculum.

14S: Committed Action The “Committed Action” program was designed to teach the relations of sameness, comparison, opposition, difference, and perspective taking. This program consisted of four steps, whereby participants were initially taught to identify a value, then identify a barrier in opposition to this value. The third step required participants to state a committed action in opposition to the barrier. For example, after being asked by the clinician, “What value means the most to you?” the participant answered “friendship.” In this case, the participant was then asked to state the barrier that is the opposite of friendship (e.g., being mean to friends) and what a committed action would be that was in line with that value (e.g., be nice to friends and hang out with them). After participants reached a predetermined criterion in terms of independent correct responding, the clinician moved on to the fourth step. The clinician presented a scenario that happened in the past and asked the client to identify whether it was in line with their values (e.g., “Last week someone told you a rumor about Bobby, and you told Max.”). If it was not in line with their values, the clinician then asked the client to list what they could do now so the behavior conforms to their values. This was done to assess whether the client could generalize the skill to novel settings. No feedback was provided during this step.

14T: Defusion The “Defusion” program was designed to teach the relations of sameness, comparison, opposition, and hierarchy. This program consisted of five steps. In Step 1, participants were taught to identify a painful thought (e.g., “I’m bad at math.”). In Step 2, the clinician described an object and

asked participants to describe how that painful thought was the same or opposite of that object (e.g., “How are the thought ‘I’m bad at math’ and a triangle the same?”). In Step 3, the clinician described another object and asked the participant to describe how the painful thought belonged to the object, or how the object belonged to the painful thought (e.g., “How does ‘I’m bad at math’ belong to the dragon?”). In Step 4, the clinician described a different object and asked the participant to describe how the painful thought was better or worse than the object (e.g., “How is ‘I’m bad at math’ worse than a broken arm?”). In the previous example, the participant may respond,

The thought is the same as a triangle because they have rough edges, and it hurts when I touch them. The thought belongs to the dragon because it is not real, and the thought is just something I have in my mind. The thought is better than a broken arm because I could recover from the thought quicker.

Once participants reached a predetermined criterion in terms of independent correct responding, the clinician moved on to Step 5. In Step 5, the clinician asked the client to identify on a scale of one item (e.g., a triangle) to another (e.g., a dragon) where the painful thought was located and what that meant (e.g., the participant may respond, “It is more like a dragon because it is just pretended.”). No feedback was provided in the last step.

14U: Defusion With Art The “Defusion With Art” program was designed to teach the relations of sameness and difference. This program consisted of five steps. In Step 1, participants were taught to create a bad thought out of art materials. In Step 2, participants were taught to write down a good thought they had. In Step 3, participants were taught to identify how those two items were the same and how they were different. In Step 4, participants were taught to take the art representing the bad thought made earlier and change it to represent the good thought. For example, the participant made an angry clay monster in Step 1 and wrote down “I am a good helper” in Step 2. The clinician asked how the two items were the same and different. The clinician then asked the participant to make the clay monster into a clay helper. Once participants reached a predetermined criterion in terms of independent correct responding, the clinician moved on to Step 5. In Step 5, the clinician asked the client to expressively identify their private events after the change in their art creation. In the previous example, the clinician may ask, “What happened to your thoughts when you changed the angry clay monster to a good helper?” The participant may respond with “The angry clay monster could turn into something better.” No feedback was provided in the last step.

14V: Self-as-Context The “Self-as-Context” program was designed to teach the relations of sameness, difference, and perspective-taking. This program consisted of five steps. In Step 1, participants were taught to write down three things about themselves in the present, past, and future. In Step 2, participants were taught to describe how statements about themselves in the present and the past were the same or different. In Step 3, participants were taught to describe how statements about themselves in the present and the future were the same or different. In Step 4, participants were taught to match each statement to an arbitrary symbol (e.g., “ Ω ,” “ $\mathbb{M}$,” “ $\times$ ”). Once participants reached a predetermined criterion in terms of independent correct responding, the clinician moved on to Step 5. In Step 5, the clinician presented an empty three-circle Venn diagram with the labels “present,” “past,” and “future.” Participants were asked to put the arbitrary symbols used in Step 4 into the appropriate circles. No feedback was provided in the last step.

14W: Values, Goals, and Actions The “Values, Goals, and Actions” program was designed to teach the relations of sameness and hierarchy. This program consisted of four steps. In Step 1, participants were taught to sort index cards describing different behaviors based on what values they aligned with. In Step 2, participants were taught to match those index cards with pictures of coordinated actions. Step 3 involved teaching participants to match pictures of actions with arbitrary symbols. For example, the clinician placed three index cards on the table that read “spend time together,” “dine at home,” and “exercise.” The clinician then placed two index cards on the table describing values: “family” and “health.” Participants were taught to sort the index cards describing behaviors based on values. Then, participants were taught to match these three index cards to pictures of someone engaging in that behavior. Participants were also taught to match these pictures to arbitrary symbols (e.g., a picture of a family eating together is the same as “ $\odot$,” a picture of a person running is the same as “ $\mathcal{O}$ ”). After participants reached a predetermined criterion in terms of independent correct responding, the clinician moved on to Step 4, where they presented a flag outline to the participant. Participants were asked to draw a flag using the symbols they learned in Step 3 that corresponded to a specific value. For example, the participant would draw “ $\mathcal{O}$ ” on the flag for the value of health. No feedback was provided during this step.

14X: Values Perspectives The “Values Perspectives” program was designed to teach the relations of sameness, opposition, and perspective-taking. This program consisted of four steps. In Step 1, participants were taught to create a list of their values. In Step 2, the clinician described one behavior and taught the client to identify which value from the list was the opposite of the behavior. In Step 3, the clinician described one behavior and taught the client to identify which value from the list aligned with the behavior. For example, the participant

wrote down “healthy” and “honesty” in Step 1. In Step 2, the clinician asked, “What is the opposite of eating a bag of chips?” In Step 3, the clinician asked, “What value does eating fruit align with?” Once participants reached a predetermined criterion in terms of independent correct responding, the clinician moved on to Step 4. In Step 4, the clinician described one behavior of another person and provided a perspective-reversal statement. Participants were asked to identify the value aligned with that behavior (e.g., “A person told the truth when they cheated on a test. If that person was you and you were that person, what is your value?”). No feedback was provided in the last step.

14Y: Acceptance The “Acceptance” program was designed to teach the relations of sameness, opposition, and difference. The program consisted of four steps. In Step 1, the clinician presented an array of symbols and images and asked the participant to select one that represented themselves. In Step 2, the participant was taught to describe how they were the opposite of the symbols they chose. In Step 3, the participant was asked to choose between not being the symbol at all and sometimes being the opposite of the symbol. The participant was taught to accept the opposite in this step. For example, the participant chose the sun as a symbol for themselves in Step 1. In Step 2, the participant said they were the opposite of the sun because sometimes they were not shiny. In Step 3, the clinician asked, “Would you rather be the sun and sometimes not be shiny, or not be the sun at all?” Once the participant reached a predetermined criterion in terms of independent correct responding, the clinician moved on to Step 4. In Step 4, the clinician asked the participant to identify one thought and describe something good and something bad about the thought. The clinician then asked whether the participant would rather have the thought sometimes and experience the good and the bad, or not have the thought at all. No feedback was provided in the last step.

14Z: I Contact The “I Contact” program was designed to teach the relations of sameness, comparison, and perspective-taking. The program consisted of five steps. In Step 1, participants were taught to draw a picture of things they had done in the past. In Step 2, participants were taught to draw a picture of things they would do in the future. In Step 3, participants were taught to draw a picture of things they were doing in the present. In Step 4, the clinician presented each of the pictures to participants and taught them to act out the feelings from the picture. Once participants reached a predetermined criterion in terms of independent correct responding, the clinician moved on to Step 5. In Step 5, the clinician asked the participant to perform a challenging task (e.g., walking with a book on their head) while thinking about the feeling from one of the pictures. The clinician repeated this with the remaining two pictures and asked participants to identify when it was the easiest to do their best. No feedback was provided in the last step.

Table 2 Summary of the relational training protocol

Program name	Psychological flexibility process targeted
14S: Committed Action	Committed action
14T: Defusion	Defusion
14U: Defusion With Art	Defusion
14V: Self-as-Context	Self-as-context
14W: Values, Goals, and Actions	Committed action
14X: Values Perspectives	Values
14Y: Acceptance	Acceptance
14Z: I Contact	Present-moment awareness

Only one program was introduced to each participant at a time. The next program was introduced once the participant reached the mastery criterion, which was a score above 90 for three consecutive trial blocks for all steps of the program. Table 2 provides an overview of these eight relational training programs. The detailed protocol is available upon request from the corresponding author.

Measures

Maladaptive Behaviors

The primary dependent variable was the frequency of participants' maladaptive behavior. The participants' maladaptive behavior was individually defined, and the clinician used a data sheet to tally each occurrence of these behaviors during participants' ACT sessions. Maladaptive behaviors that occurred outside the ACT-based instruction were not recorded. For Briana, her maladaptive behavior included improper sitting and unresponsiveness to questions. Improper sitting was defined as sitting in a chair and leaning back so that her head was not upright, leaning forward so that her head was on the table, leaning on her body parts, or removing her body from the chair during instructions. Unresponsiveness to questions was defined as the refusal or failure to respond within 7 s of a question being asked or providing a response that was unrelated to the question. For Josie, her maladaptive behavior included distractibility, which was defined as any instance of being unresponsive to a question asked, having off-topic conversations outside of break time, or playing during instructions. Tina's maladaptive behavior included fidgeting, which was defined as her holding an item or her body part in her hand and engaging in repetitive movements.

ACT Quantitative Analysis Scale

A secondary dependent measure was collected in order to directly assess the effect of the relational training on ACT sessions. Participants' engagement during ACT sessions was

measured using the ACT Quantitative Analysis Scale (AQAS; Kryszak et al., 2017). The scale used three dimensions to evaluate participants' engagement during each ACT session: precision, scope, and depth.

Precision was evaluated by the extent to which participants' responses were detailed and concise, met the activity requirement, and fully answered the question. Engagement with low precision consisted of answers that were not applicable or considered broad and generic. Examples of low-precision responses included the participant saying, "I can help my family more often" when asked about committed actions, or providing the response, "I won't get upset if I don't get my way" when asked about times they may practice acceptance. Engagement with high precision consisted of logically sound, detailed answers that fully answered the question being asked. Examples of high-precision responses included a participant saying, "This week I will independently complete my daily chores without my mom having to remind me" when asked about committed actions, or responding with "I get so upset when I lose a game that I miss out on having fun during the next game" when asked about situations in which they could benefit by practicing acceptance.

Scope evaluated the extent to which participants' responses fulfilled the activity requirement and translated easily into novel areas related to the original question. Engagement with low scope consisted of answers that were not applicable or applied only to the current activity or discussion. Examples of low-scope responses included the participant saying, "I'm really hungry right now" when asked to identify thoughts and feelings that make it difficult to stay focused on the present moment, or responding with "I get to pick what I want to work for" when asked about the importance of values. Engagement with high levels of scope consisted of answers that met the requirements of the current activity and were also elaborated on to fit novel situations. Examples of high-scope responses included a participant saying, "Whenever we meditate, and during class, I notice that my mind usually wanders off and I can't stop thinking about all my homework and chores that I have to do once I get home" when asked about thoughts that interfere with the present moment, or responding with "Focusing on my value of education helps me stay on track and reminds me why getting my homework done is so important" when asked about the importance of values.

My thoughts are like balls in a pool because they float to the top, and the more I try to push them down and make them go away, the more I struggle, even when struggling is more work than just letting them be there and eventually float off.

Clinicians used a 4-point scale to rate the participant's response in each dimension, and the subscores were summed to form a total score. AQAS scores could range from a minimum of 0 to a maximum of 9. The clinician used the AQAS to rate the participant's engagement in ACT sessions at the end of each session.

Procedures

The current study used a concurrent multiple-baseline across-participants design to evaluate the effect of relational training on participants' engagement in ACT sessions in order to decrease maladaptive behavior. During baseline, participants first completed the ACT-based intervention before returning to their regular ABA session. During the intervention, participants' regular ABA sessions were replaced by the relational training, followed by the ACT-based intervention. Once the participant mastered all eight programs, maintenance probes were conducted. No consequences were provided for maladaptive behavior throughout the study.

Interobserver Agreement and Procedural Fidelity

The interobserver agreement (IOA) for the primary dependent variable was collected for 38% of the sessions. A second observer, who was a master's-level graduate student enrolled in a behavior-analytic program working at the same clinic, recorded the frequency of the maladaptive behavior. The second observer used a data sheet to tally the participant's maladaptive behavior during ACT sessions. The data sheet also contained the operational definition of the target behavior. During each session, the total-count IOA was calculated by dividing the smaller frequency observed by the larger frequency observed and multiplying by 100%. The average total-count IOA was 87.84% (range 33.33%–100%, $SD = 19.68\%$).

Procedural fidelity was calculated by the first author, who reviewed all recordings of the session and used a checklist to examine whether the therapist followed the session structure described previously, delivered the relational training as instructed, and recorded data in a timely manner. There were 10 items in total on the fidelity checklist, and the procedural fidelity was calculated by dividing the number of items the therapist completed by 10 and multiplying by 100%. The procedural fidelity was 100% for all sessions.

Results

During the intervention phase, all three participants showed improvement in their performance during the relational training. Both Josie and Tina mastered all eight relational training programs. Briana left the study early due to a family emergency. At exit, she had mastered six programs. Table 3 displays the session number when each program was mastered by the participant.

Figure 1 displays participants' frequency of maladaptive behavior and their engagement during ACT sessions. Overall, the data showed that ACT was ineffective in terms of decreasing maladaptive behavior during baseline. However, as participants progressed through the relational training, Briana showed a decreasing trend in her maladaptive

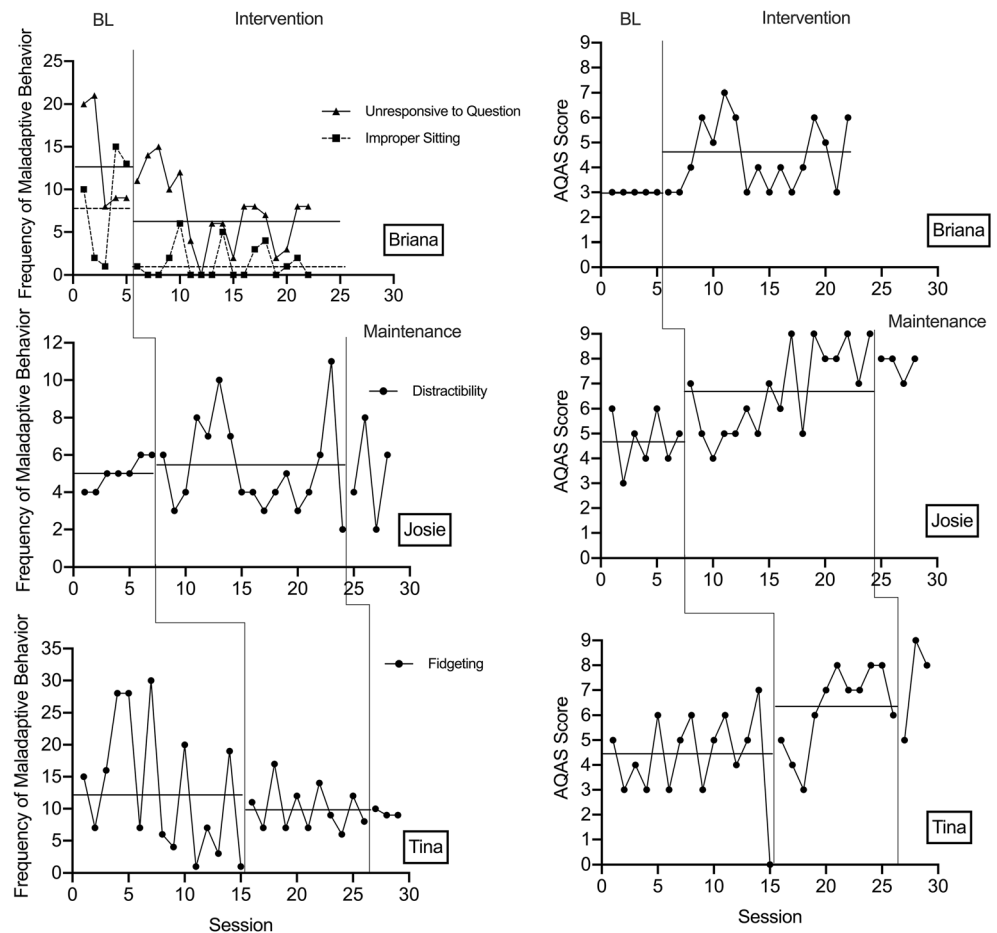
Table 3 Summary of the relational training outcome

Session no.	Relational training program mastered during this session		
	Briana	Josie	Tina
10		10Z	
12		14X	
13	14Z		
15		14Y	
16	14X		14Z
17		14U	
18	14U	14T	14Y
19	14S	14S	
20		14V	14U
21			14S
22	14V, 14Y		14X
23		14W	
24			14W
25			14T
26			14V

behavior, and the variability and level of Tina's maladaptive behavior also decreased. All three participants' AQAS scores showed various levels of improvement after the introduction of the relational training protocol when compared with their performance during baseline.

Specifically, Briana's frequency of maladaptive behavior was at a variable, high level during baseline, with an average of 13.4 instances of unresponsiveness to questions and 8.2 instances of improper sitting each session. Following the introduction of the relational training, their frequency decreased in both level and variability. The Cohen's d for the two maladaptive behaviors was 0.57 and 1.25, respectively, indicating a medium to large treatment effect. Josie's frequency of maladaptive behavior showed an increasing trend during baseline, with an average of five instances of being distracted each session. Following the introduction of the relational training, the frequency increased in variability with no change in level ($M = 5.35$). A similar pattern was observed during the maintenance phase ($M = 5$). The Cohen's d for Josie's behavior was 0.06, indicating no treatment effect. Tina's frequency of maladaptive behavior showed a variable decreasing trend during baseline, with an average of 12.8 instances of fidgeting each session ($SD = 10.22$). Following the introduction of the relational training, although the frequency continued showing a decreasing trend with a significant decrease in variability ($M = 10$, $SD = 3.49$), the Cohen's d was 0.15, indicating no treatment effect in decreasing the level of her fidgeting behavior. This decreased variability was maintained during the maintenance phase ($M = 9.33$, $SD = 0.58$).

Fig. 1 The frequency of maladaptive behavior and participants' engagement during acceptance and commitment training (ACT) sessions. *Note.* The horizontal line shows the average of dependent variables during the baseline and the intervention phases. BL = baseline; AQAS = ACT Quantitative Analysis Scale



In terms of the effect of the relational training on participants' engagement during ACT sessions, Briana's scores remained stable at three points during the baseline condition. During the intervention, her engagement during ACT sessions ranged from 3 to 7, with an average of 4.41. The percentage of nonoverlapping data (PND) was 64.71%, indicating a small to moderate treatment effect. A family emergency occurred between Session 21 and Session 22, and she ended her participation before mastering all eight relational training programs. Josie's engagement during ACT sessions averaged 4.71 during baseline. During the intervention, her AQAS score showed an increasing trend. Her score ranged from 4 to 9, with an average of 6.71. The PND was 52.94%, indicating a small to moderate treatment effect. She maintained a higher level of engagement in ACT sessions during the maintenance probe. Tina's engagement during ACT sessions averaged 4.33 during baseline. During the intervention, her AQAS score showed an overall increasing trend, ranging from 3 to 8, with an average of 6.27. The PND was 27.27%, indicating a small treatment effect. Her improved engagement during ACT sessions was maintained during maintenance probes. The Cohen's *d* measures for participants' AQAS scores were 0.31, 0.30, and 0.34, indicating a small treatment effect.

Discussion

The current study demonstrates preliminary evidence for the overall effectiveness of using relational training to improve the effect of ACT-based instructions. Visual analysis indicated that the frequency of maladaptive behavior did not decrease during baseline when ACT was implemented as a stand-alone treatment. After the introduction of the relational training, one participant showed a decrease in maladaptive behavior, and another participant showed a decrease in the maladaptive behavior's variability. All participants improved their engagement during ACT sessions after the introduction of relational training programs that targeted various high-order operants of relational framing. By using relational trainings to increase the complexity of individuals' abilities to engage in arbitrarily applicable relational responding, the current study proposed a promising solution that aids ACT-based interventions with individuals diagnosed with autism. The use of the AQAS also provided ways to evaluate the engagement during ACT sessions in a quantifiable manner without relying on participants' self-reported psychological flexibility measures.

There are several potential explanations as to why ACT as a stand-alone intervention failed to produce a significant

decrease in participants' maladaptive behavior. Individuals diagnosed with autism often show deficits in various relational frames, and these deficits have been found to be significantly related to language abilities (Devany et al., 1986; Kent et al., 2017; Rehfeldt et al., 2007). ACT protocols often involve experiential exercises, metaphors, and discussion of private events that are abstract and covert (Luoma et al., 2007), and the language-based intervention could be less effective if individuals cannot reliably understand the clinician. What we define as "understanding" will clearly require subsequent investigations, yet our work here suggests that relational framing abilities may be at least part of this analysis.

Another reason lies in individuals' abilities to actively engage in derived relational responding. Although the individual may be able to complete the activity as required, the inability to engage in arbitrarily applicable relational responding may prevent the transfer of the stimulus functions, resulting in a less desirable treatment outcome. Individuals diagnosed with autism may require additional prompting during experiential exercises and discussion (Szabo, 2019). Although they may be able to verbally state the correct answer and demonstrate comprehension of the activity at a later time, there is the risk that such response is contingency shaped in nature. In other words, their response could be the result of a history of direct reinforcements, not the product of derived relational responding.

The relational training provides two unique benefits in this circumstance. First, it improves clients' ability to engage in relational framing (Daar et al., 2015; Dixon et al., 2019). This will further develop clients' language ability and help them better understand the ACT language. Second, each of the eight relational training programs used in this study involved repeated practice of behavior that involved responding across multiple relational frames. By targeting behavior that is highly related to processes of the psychological flexibility model and by using multiple sets of stimuli in each program, these eight programs can be seen as multiple-exemplar trainings on the six processes. Previous research has shown the effectiveness of multiple-exemplar training procedures in producing changes across various contexts (Marzullo-Kerth et al., 2011; Rosales et al., 2011). It is possible that this procedure produced a similar outcome by promoting the emergence of derived relations that are in line with the psychological flexibility model. This is further supported by the secondary AQAS measure, as all three participants showed an increase in scores after the introduction of the relational training. It could be the case that participants failed to benefit from ACT sessions during baseline because they could not actively participate in them.

The current study is the first to explore the effect of a dedicated relational training protocol on the effect of ACT interventions with children diagnosed with autism. The current finding has several implications for practitioners. First of all, the result can be interpreted as preliminary evidence that supports relational training as a potential remedy for

individuals diagnosed with autism who are not responding to ACT interventions as expected. Their underlying deficits in language and relational responding may prevent them from benefiting from ACT to its full potential, and relational training seems to address this issue.

The second implication lies in the format of the relational training. Two of the three participants mastered all eight programs during the study. Briana had mastered six of the eight programs by the time she exited the study due to a family emergency. This shows that behavior analysts who are less comfortable implementing language-based interventions can adapt ACT protocols and approach the intervention using formats that they are more familiar with. Programs in the relational training protocol each corresponded to one process of the psychological flexibility model and were carried out in a discrete-trial training format. By tracking changes in observable behavior and using common techniques in behavior-analytic practices (e.g., prompting, prompt fading), practitioners can use advancements in ACT to benefit their clients while continuing to use techniques they are familiar with. This can be helpful when the funding source restricts the format of treatment, or when direct implementers are not familiar with the ACT language. The current study provides preliminary evidence that ACT-based interventions can take a form that is more aligned with the traditional format of behavior-analytic services.

Furthermore, the current study can be seen as preliminary direct evidence that some level of competency in relational responding is needed in order for ACT-based interventions to be effective. Participants in the current study all showed improvements in derived relational responding during the relational training, and concurrently, ACT began to show its effectiveness at decreasing maladaptive behavior. We also observed an increase in engagement during ACT sessions. The relationship between relational training and improved engagement was demonstrated using a multiple-baseline across-participants design. However, limited by the sample size and the nonstandardized baseline measures of participants' ability to engage in relational responding, we did not conduct a parametric analysis on the amount of improvement made between the relational training and ACT sessions. As future studies evaluate and establish standardized measures of relational framing and ACT outcome, it would be interesting to identify a clinical cutoff criterion on which practitioners can base an informed decision on whether ACT-based interventions are likely to be effective.

Last but not least, the current study used the AQAS to capture participants' engagement during the ACT session. This provides an alternative way practitioners might capture ACT intervention outcomes. Because measures of psychological flexibility cannot be taken during each session, and inexperienced implementers may have a hard time agreeing on traditional measures of ACT outcomes (e.g., identifying

instances of flexible and inflexible processes), the AQAS provides an easy way for practitioners to track client progress session to session.

Despite the promising outcome, the current study should be interpreted with caution. The frequency of maladaptive behavior during ACT sessions was used as the indicator of socially significant changes caused by the intervention. It is possible that the observed change is a result of repeated exposure to ACT, rather than an effect of improved participation during ACT sessions as a result of the relational training. Although the concurrent multiple-baseline design decreases the likelihood that this potential confounding variable is the cause of the observed change, this conclusion should be made with caution, given the small sample size and the fact that the third participant already showed a decreasing trend in maladaptive behavior before the introduction of the relational training. Nonetheless, the current data show that the relational training has some effect on the effectiveness of ACT, in terms of both the primary dependent variable (i.e., decrease in maladaptive behavior) and the secondary dependent variable (i.e., improved engagement during ACT sessions). It should be pointed out, though, that this conclusion may not generalize to ACT's utilities besides what has been demonstrated in the current research, in terms of decreasing the frequency and variability of the maladaptive behavior and improving engagement with ACT sessions. More data are needed to examine the relationship between relational training and the effectiveness of ACT. Future studies can expand this study's external validity by replicating the current procedure with more participants and tracking changes in social-emotional challenges outside of the clinical setting. Future studies could also measure participants' repertoires in relational responding and compare their level before and after the intervention to identify potential relationships between the ability to engage in arbitrarily applicable relational responding and improved engagement in ACT sessions. Future researchers can conduct component analyses to pinpoint the most effective part of the current treatment package.

It should also be pointed out that the current study is preliminary in nature. It contains a small group of participants, and the measures obtained during baseline are variable. However, because high variability often underestimates the effect size, the current dataset allowed a more conservative interpretation of the relational training's effect on ACT sessions. Future research could replicate the current procedure with individuals with various degrees of delay in language. Although the current data provide evidence that the relational training protocol improved participants' engagement during ACT sessions, two of the three participants failed to show clear decreases in their maladaptive behavior. A longer exposure to ACT, a strengthened relational training protocol, and expanded stimuli during relational training may improve the overall effectiveness of the treatment package. It is worth

noting that the observed relationship between the relational training protocol and ACT's effectiveness in decreasing maladaptive behavior is less clear, as the only participant who showed a decrease in the level of maladaptive behavior was the one who exited the study before completing the relational training protocol. Future research should expand this preliminary finding by increasing the number of participants and by continuing the intervention until the primary dependent variable has been reduced to a socially significant level.

The third limitation lies in the use of the AQAS as the secondary measure of engagement during ACT sessions. Although the AQAS provides an operational definition for each score across all three dimensions, the scoring can be affected by the rater's bias. The use of Likert-type scales in the AQAS still relies on the clinician's subjective rating of participants' behavior. The lack of IOA and the fact that the rater was not blind to the phase that the participant was in undermine its validity. However, it should be pointed out that we include the AQAS here as a secondary measure to document changes in the participants' participation in ACT. Most often, research documents ACT's effectiveness by examining the clinical outcome (Ruiz & Roche, 2007). Scales measuring psychological flexibility rely heavily on participants' self-report (e.g., the Acceptance and Action Questionnaire, the Avoidance and Fusion Questionnaire for Youth), and research has yet to reach a conclusion on whether self-reported measures can yield reliable results with individuals diagnosed with autism (Bishop & Seltzer, 2012; Ozsivadjian et al., 2014). By using the AQAS, we attempted to provide a somewhat objective measure to capture participants' responsiveness to the ACT protocol. Future research should examine whether the AQAS provides an accurate rating of the client's participation, or develop a more precise tool that practitioners can use to determine whether the client is responding to ACT appropriately.

In conclusion, the current data provide initial support for a method to improve the effectiveness of ACT when attempting delivery to children diagnosed with autism who have underdeveloped language repertoires. A universal application of ACT to clients may come with mixed results, and these data serve as a cautionary tale. Unless the underlying relational framing repertoires are first established for a client, it becomes challenging to successfully produce comprehension of the metaphorical nature of the therapeutic processes that ACT rests upon. Furthermore, these data demonstrate a long-overdue empirical linking of relational frame theory's basic processes with the clinical application of those processes in ACT treatments. As behavior analysts providing care for persons with disabilities increasingly seek treatment options that focus on altering language and its effects on overt behavior, continued research in the area of ACT interventions holds great potential to produce positive changes for the human condition.

Declarations

Conflicts of interest Mark R. Dixon receives small royalties from sales of PEAK and AIM. The remaining authors declare no conflicts of interest.

Consent to participate Written consent was obtained from all participants' guardians.

Ethics approval All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional research committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards. The study was approved by the Human Subject Committee of Southern Illinois University.

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